

PART A – Data Exploration

Q1 Understanding Data Types and Transformations

Q1.1 Classification of variables according to measurement type

A critical first step in predictive analytics is to understand the structure and measurement types of the variables within the dataset (McCarthy et al., 2022). This ensures that appropriate data preparation techniques can be applied before building predictive models. Using RStudio functions such as *str()* and *glimpse()*, the ConnectTel dataset was examined and found to contain 10,500 customer observations and 38 variables, consisting of a mixture of numerical and categorical variables. The variables were classified into continuous numerical, ordinal, nominal, and binary indicator categories based on how the data is measured and how it should be interpreted for modelling purposes.

Continuous numerical variables

Continuous numerical variables represent measurable quantities where values exist on a continuous scale (Uher, 2021). These variables typically describe customer usage behaviour, financial contributions, or calculated engagement metrics. The continuous variables identified in the dataset include network tenure, data usage metrics, voice usage statistics, service counts, engagement scores, and the target variable average monthly bill amount.

Examples include:

- **network_stay**, which measures the length of time a customer has remained with the provider
- **dusage_sum**, **dusage_avg**, and **total_data_activity**, which measure customer data consumption behaviour
- **vusage_onnet_sum** and **vusage_offnet_sum**, which measure voice call usage patterns
- **add_on_tot_rental**, which represents the financial contribution of subscribed services
- **digital_engagement_score**, which reflects customer interaction intensity
- **avg_monthly_bill_amount**, which represents the target variable for prediction

These variables are classified as continuous because they measure quantities that can take a wide range of values and where differences between values are meaningful. For example, a customer using 3000 MB of data clearly consumes more than a customer using 1000 MB, and this difference has analytical significance. Such variables are particularly suitable for regression modelling because they preserve magnitude and variance information necessary for prediction.

Ordinal variables

Ordinal variables represent categorical data with a natural ranking but without equal spacing between categories (Lalla, 2016). The key ordinal variable identified in the dataset is: **age_group**

This variable contains ordered categories such as: <20, 20–30, 30–40, 40–50, 50–60, 60–70, and 70+.

This variable is considered ordinal because it represents increasing age ranges. While the categories are ranked, the exact numerical distance between categories cannot be assumed to be equal (Lalla, 2016). For example, the behavioural difference between customers aged 20–30 and 30–40 may not be identical to that between 50–60 and 60–70. Proper handling of ordinal variables is important because treating them as

nominal would ignore valuable ordering information, while treating them as continuous may introduce incorrect assumptions (Robitzsch, 2020).

Nominal variables

Nominal variables represent categorical classifications where no natural ordering exists (Anderson, 1984). The nominal variables identified include `device_type`, `device_category`, `district_name` and `gender`. These variables represent customer characteristics rather than measurable quantities. For example, the `device_type` has categories which including 2G, 3G and 4G. These categories represent technology types rather than increasing numerical levels. Although 4G is technologically more advanced than 3G, the numerical difference between categories does not represent a measurable quantity suitable for regression analysis.

Similarly, `device_category` also has Smartphone, Feature phone, Tablet, Modem and Basic, which represent device classification rather than ranked categories.

`district_name` represents geographic location and contains multiple regional categories. Since location labels do not have numerical meaning, they must be encoded before modelling. Nominal variables cannot be directly used in predictive modelling because algorithms interpret numbers as magnitude relationships. Therefore, appropriate encoding techniques must be applied (Baladram, 2025).

Binary indicator variables

Binary variables contain only two possible outcomes and are often used to represent presence or absence of a characteristic (Modelling Binary Data, Second Edition, n.d.). The dataset contains one naturally binary variable which is `gender` and this variable contains two categories: Male and Female. This variable can be transformed into a binary indicator variable (0/1) for modelling purposes. Binary variables are useful in predictive modelling because they allow models to estimate the impact of categorical differences (Fenlon et al., 2017).

Q1.2 Variables that may not contribute meaningful information for predictive modelling

Not all variables in a dataset necessarily contribute useful information for predictive modelling. Some variables may introduce noise, redundancy, or unnecessary complexity, which can reduce model performance and interpretability (Biecek & Burzykowski, 2021).

Customer identifier variable

The variable **`customer_id`** represents a unique identifier assigned to each customer. This variable does not provide useful predictive information. First, it contains no behavioural or demographic information related to billing patterns. Second, each value is unique, meaning no generalizable patterns can be learned. Third, including such identifiers may cause models to overfit meaningless patterns. Therefore, this variable should be removed prior to modelling. Removing identifier variables is standard practice in predictive analytics to ensure that only meaningful features are included.

Redundant usage variables

The dataset also contains multiple derived variables measuring similar behaviours. For example:

- `total_data_activity` is closely related to `usage_sum`
- `avg_data_per_day` is derived from data usage patterns
- `total_voice_usage` overlaps with voice usage components

The correlation matrix produced in R indicates that several of these variables exhibit strong relationships. While such variables individually may be informative, including all of them simultaneously may introduce multicollinearity (Kim, 2019). Multicollinearity can reduce model stability, increase coefficient variance and reduce interpretability of regression results (Kim, 2019). To address this issue, feature selection techniques such as correlation filtering or dimensionality reduction should be applied later in the modelling process.

Variables containing missing values

Initial inspection also identified several variables containing missing values, including: *usage_onnet_avg*, *usage_offnet_avg*, *add_on_tot_rental*, *add_on_intensity* and *digital_engagement_score*. These variables cannot be directly used in predictive models because most algorithms cannot process missing values without preprocessing (Alexandropoulos et al., 2019). Missing data may also distort statistical distributions. Appropriate handling techniques such as imputation or removal must be applied. Therefore, this issue will be explored further in the missing data analysis section.

Q1.3 Appropriate transformation techniques for categorical variables

Since predictive models such as regression and decision trees require numerical input, categorical variables must be transformed into suitable numerical representations (Loh, 2011).

Factor conversion

The first step involves converting categorical variables into factor data types within R. This ensures the software correctly interprets the variables as categorical rather than text data

One-hot encoding

Nominal categorical variables were transformed using dummy encoding to convert categories into binary indicators. This allows regression models to interpret categorical effects without imposing artificial ranking. For example, the *device_type* variable was transformed into separate variables such as *device_type_3G* and *device_type_4G*. This transformation ensures that each category is treated independently.

Ordinal encoding

For ordinal variables such as *age_group*, ordered factor encoding was applied to preserve ranking information. Therefore, this allows models to recognize that higher age categories represent progression rather than unrelated categories.

Feature scaling

Although decision tree models do not require scaling, regression models benefit from normalization or standardisation of continuous variables (Frank et al., 1998). Scaling ensures variables with large magnitudes do not dominate model learning. Some of the scaling methods that can be used are Min-max normalization and Z-Score standardization (Frank et al., 1998). This step may be applied later depending on model requirements.

Q2 Exploratory Data Analysis

Q2.1 Structured data preparation workflow

Before developing predictive models, it is essential to conduct a structured data preparation workflow to ensure the dataset is accurate, complete, and suitable for modelling (Sand Technologies, 2025). A

systematic preparation process reduces modelling errors and improves predictive reliability. Based on the exploratory analysis conducted in RStudio, the following workflow is recommended.

Data structure verification

The first step involved examining the structure of the dataset using functions such as *str()* and *glimpse()* to confirm variable types and detect inconsistencies. This step is necessary because incorrect data types may cause modelling errors. For example, categorical variables stored as character data must be converted into factors before modelling. This step ensures that numerical variables are correctly recognised and categorical variables are properly encoded.

Missing value identification

The next step involved identifying missing values across all variables. The missing value analysis revealed that several variables contain substantial missing data.

Key findings include below and output at the Appendix.

- vusage_onnet_avg contains approximately 82.6% missing values
- vusage_offnet_avg contains approximately 31.6% missing values
- add_on_tot_rental, add_on_count, and add_on_intensity each contain approximately 36.5% missing values
- digital_engagement_score contains approximately 38.2% missing values
- dusage_avg and total_data_activity contain about 8.2% missing values
- Small missing proportions were observed in device_type, device_category, gender, and age_group

This step is important because missing data can significantly affect model accuracy. Many predictive algorithms cannot process missing values directly and may either fail or produce biased results. Therefore, missing data must be handled through imputation or removal before modelling (Emmanuel et al., 2021).

Descriptive statistical analysis

Descriptive statistics were generated to understand the central tendency and spread of the variables. The summary statistics revealed significant variation across usage variables. According to the output has shown that the average monthly bills range from 11,446 to 19,008, with the total data activity range from 0 to 182,298, the data usage sum ranges from 0 to 176,410 and finally, the digital engagement score ranges from 1 to 7,391. Therefore, this step is necessary because it provides insight into data scale differences and highlights variables that may require normalization.

Distribution analysis

Histograms were generated to examine the distribution of all continuous variables. Most variables demonstrated strong right skewness, indicating that most customers have relatively low usage while a small number of customers exhibit extremely high usage behaviour.

Understanding distributions is important because:

- Skewed data may affect regression assumptions
- Extreme values may bias model training
- Transformations may be required

Outlier detection

Boxplots were used to identify extreme values. Many variables showed substantial outliers, particularly usage-related variables.

This step is necessary because outliers may:

- Distort regression coefficients
- Increase model variance
- Reduce prediction accuracy

Outliers must therefore be investigated rather than automatically removed.

Variability analysis

Standard deviation analysis was conducted to identify variables with the highest variability (Rakrak, 2025). This helps determine which variables contribute the most variation to the dataset and may therefore have strong predictive influence. This step is necessary because high variance variables often contain useful predictive signals but may also require scaling (Rakrak, 2025).

Correlation analysis

Correlation analysis was conducted to identify relationships between variables. Strong relationships were observed among usage variables, indicating potential redundancy (Kumar & Chong, 2018). This step is important because redundant variables may introduce multicollinearity in regression models and should be considered during feature selection.

Q2.2 Distribution and descriptive statistics analysis

The exploratory analysis revealed several important patterns regarding the distribution of continuous variables.

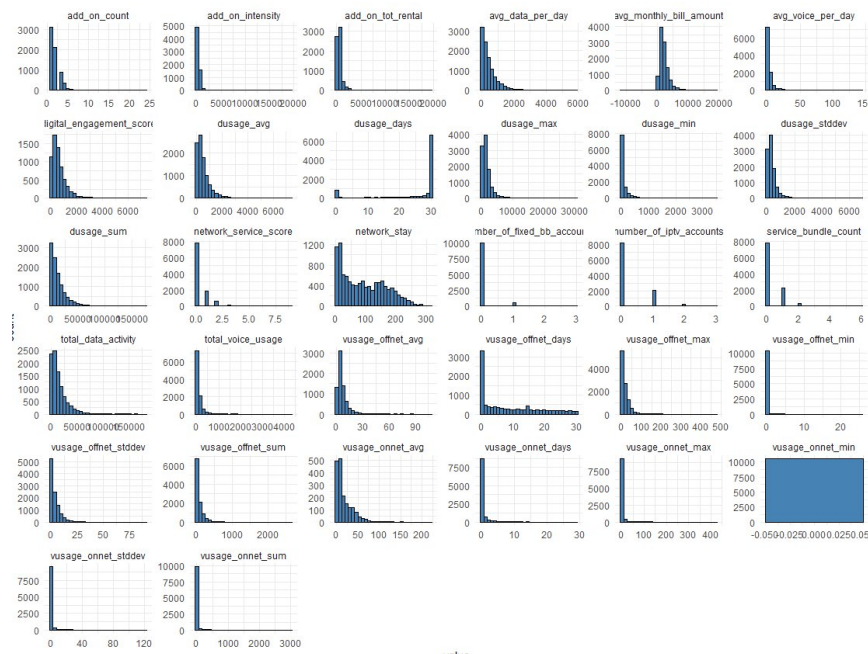


Figure 1: Histogram for all continuous variables.

The histogram analysis shows that most usage variables follow positively skewed distributions. This suggests that the majority of customers have relatively moderate usage patterns while a small group of heavy users contribute extreme values.

For example:

- **total_data_activity** shows a long right tail
- **dusage_sum** shows similar behaviour
- **total_voice_usage** also demonstrates positive skewness

This pattern is typical in telecommunications datasets where a minority of heavy users generate large usage volumes (Walelgne et al., 2020). The average monthly bill also shows moderate right skewness, suggesting that while most customers have moderate billing amounts, a small number incur significantly higher charges. Descriptive statistics further confirm these findings. The difference between median and maximum values across several variables indicates the presence of extreme observations.

Q2.3 Distributional characteristics

Variables with the largest variability

Based on the standard deviation analysis, the variables with the highest variability include: total_data_activity, dusage_sum, dusage_max, avg_monthly_bill_amount, add_on_tot_rental and digital_engagement_score. These variables exhibit large standard deviations, indicating substantial variation among customers. This suggests that customer usage behaviour differs significantly across the population, which is expected in telecommunications data where customer needs vary widely. High variability variables often provide strong predictive power because they capture behavioural differences between customers.

Variables exhibiting strong skewness

The skewness analysis revealed several variables with strong positive skewness.

```
# A tibble: 32 x 2
  variable      skewness
  <chr>      <dbl>
1 vusage_offnet_min 40.7
2 add_on_intensity 19.5
3 vusage_onnet_sum  9.86
4 add_on_tot_rental 7.18
5 vusage_onnet_stddev 6.38
6 total_voice_usage 6.20
7 avg_voice_per_day 6.20
8 vusage_onnet_max  5.78
9 vusage_offnet_sum 5.59
10 vusage_onnet_days 5.51
# i 22 more rows
# i Use `print(n = ...)` to see more rows
```

Figure 2: The skewness analysis

Strong skewness indicates that most customers have low values while a small number of customers have extremely high values. This behaviour suggests that customer usage is highly concentrated among a small number of heavy users. Highly skewed variables may require transformation such as logarithmic scaling to improve model stability.

Presence of extreme values

Boxplot analysis clearly indicates the presence of numerous extreme values across the dataset.

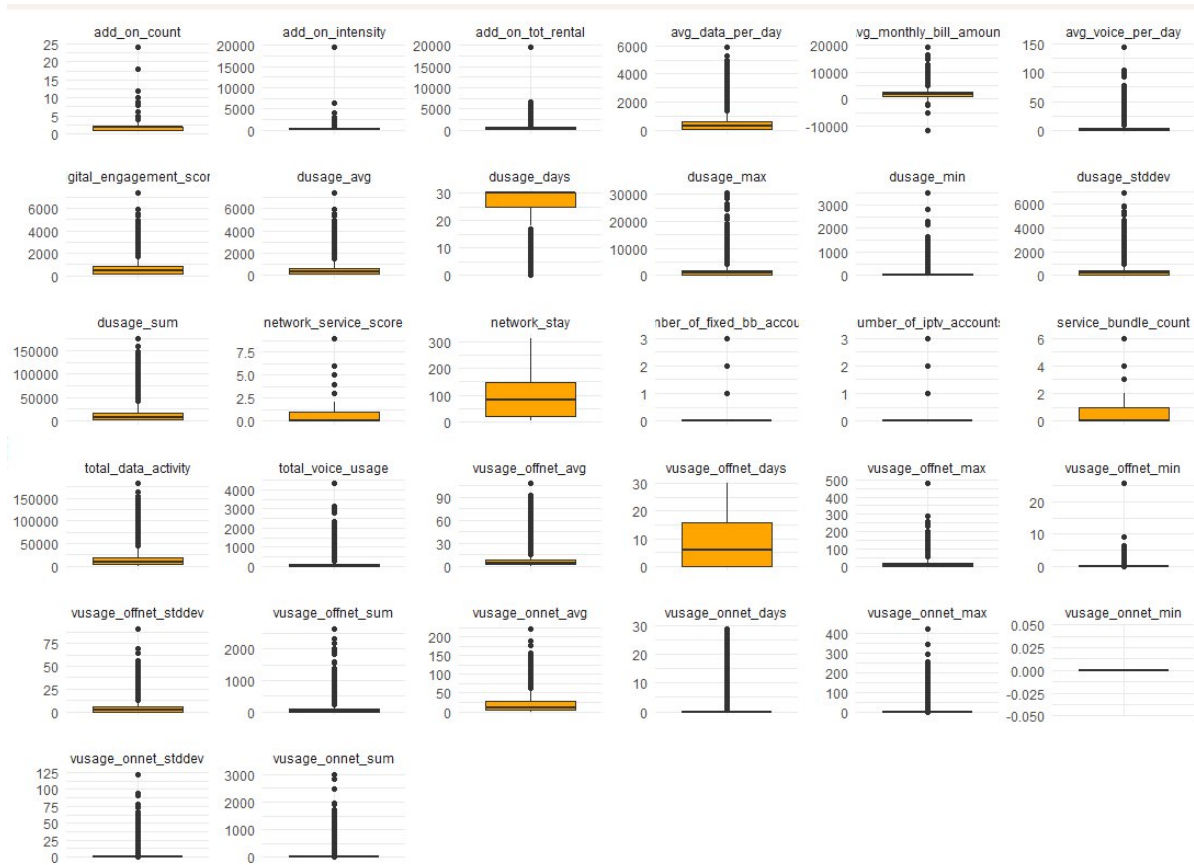


Figure 3: Box Plots for outliers detection.

Variables with large numbers of outliers include: number_of_ipiv_accounts, vusage_onnet_max, vusage_onnet_days, dusage_min, total_voice_usage, avg_voice_per_day.

The presence of these outliers suggests significant behavioural differences between customer segments. However, these extreme values are likely to represent legitimate high-usage customers rather than data errors. Therefore, careful treatment is required before any action on deletion.

Q2.4 Outlier identification and interpretation

The outlier analysis identified several variables with substantial numbers of extreme observations. According to the output, number_of_ipiv_accounts (over 2,200 outliers), vusage_onnet_max (over 1,800 outliers), dusage_min (over 1,500 outliers), total_voice_usage (nearly 1,000 outliers).

```
# A tibble: 32 x 2
  variable      outlier_Count
  <chr>      <int>
1 number_of_ipiv_accounts 2285
2 vusage_onnet_max      1828
3 vusage_onnet_days      1828
4 vusage_onnet_stddev    1828
5 dusage_days           1809
6 vusage_onnet_sum       1650
7 dusage_min            1516
8 total_voice_usage      978
9 avg_voice_per_day      978
10 vusage_offnet_sum      890
# i 22 more rows
```

Figure 4: Variables' Outliers count.

These observations likely represent heavy users rather than incorrect data entries. In telecommunications contexts, heavy users are often highly valuable customers and therefore should not automatically be

removed (Subex Limited, n.d.). Outliers may influence predictive modelling in several ways (Kalisch et al., 2016). First, they may disproportionately influence regression coefficients. Second, they may increase model variance. Third, they may reduce model generalization if not properly handled. However, removing such values may also remove important behavioural patterns (Kalisch et al., 2016).

Therefore, appropriate treatments such as Log transformation, winsorisation, robust scaling or tree-based models must be considered to implement (Kalisch et al., 2016).

Decision tree models are particularly suitable because they are less affected by extreme values compared to regression models (Czajkowski & Kretowski, 2016). Based on this analysis, it is recommended that outliers should not be automatically removed unless they are confirmed to be data errors. Instead, transformation techniques should be considered to reduce their influence while preserving valuable behavioural information.

The exploratory data analysis revealed that the ConnectTel dataset contains substantial variability, skewness, and outliers typical of telecommunications usage data. Several variables exhibit strong positive skewness, indicating that customer behaviour is concentrated among a small group of heavy users. Variables such as total data activity, data usage, and billing amounts show the greatest variability and are therefore likely to be important predictors of customer billing behaviour. Outlier analysis suggests that extreme values likely represent genuine high-usage customers rather than errors and should therefore be handled carefully.

Q3 Handling Missing Data

Q3.1 Identification of variables with missing values

Handling missing data is an important step in predictive analytics because incomplete data can bias model results, reduce statistical power, and cause algorithm failures (Emmanuel et al., 2021b). Therefore, the dataset was first examined to identify variables containing missing values. The missing value analysis revealed that several variables contain missing observations at varying levels of severity.

The analysis of missing values shows that several variables contain substantial levels of incomplete data. The most significant missing values were found in `vusage_onnet_avg`, where approximately 82.6% of the data is missing. Other variables with high levels of missing data include `digital_engagement_score` (about 38.2%), `add_on_tot_rental`, `add_on_count`, and `add_on_intensity` (each around 36.5%), as well as `vusage_offnet_avg` (approximately 31.6%).

A moderate level of missing data was observed in `dusage_avg` and `total_data_activity`, with both variables containing around 8.2% missing values. Meanwhile, only a small proportion of missing data was identified in the demographic and device-related variables, including `gender`, `age_group`, `device_type`, and `device_category`, indicating relatively good data completeness for these attributes.

```
+ arrange(desc(Missing_Percent))
```

	Variable	Missing_Count	Missing_Percent
<code>vusage_onnet_avg</code>	<code>vusage_onnet_avg</code>	8672	82.5904762
<code>digital_engagement_score</code>	<code>digital_engagement_score</code>	4010	38.1904762
<code>add_on_tot_rental</code>	<code>add_on_tot_rental</code>	3833	36.5047619
<code>add_on_count</code>	<code>add_on_count</code>	3833	36.5047619
<code>add_on_intensity</code>	<code>add_on_intensity</code>	3833	36.5047619
<code>vusage_offnet_avg</code>	<code>vusage_offnet_avg</code>	3313	31.5523810
<code>dusage_avg</code>	<code>dusage_avg</code>	858	8.1714286
<code>total_data_activity</code>	<code>total_data_activity</code>	858	8.1714286
<code>usage_intensity</code>	<code>usage_intensity</code>	858	8.1714286
<code>gender</code>	<code>gender</code>	135	1.2857143
<code>age_group</code>	<code>age_group</code>	135	1.2857143
<code>device_category</code>	<code>device_category</code>	30	0.2857143
<code>device_type</code>	<code>device_type</code>	23	0.2190476

Figure 5: Missing Value Percentage.

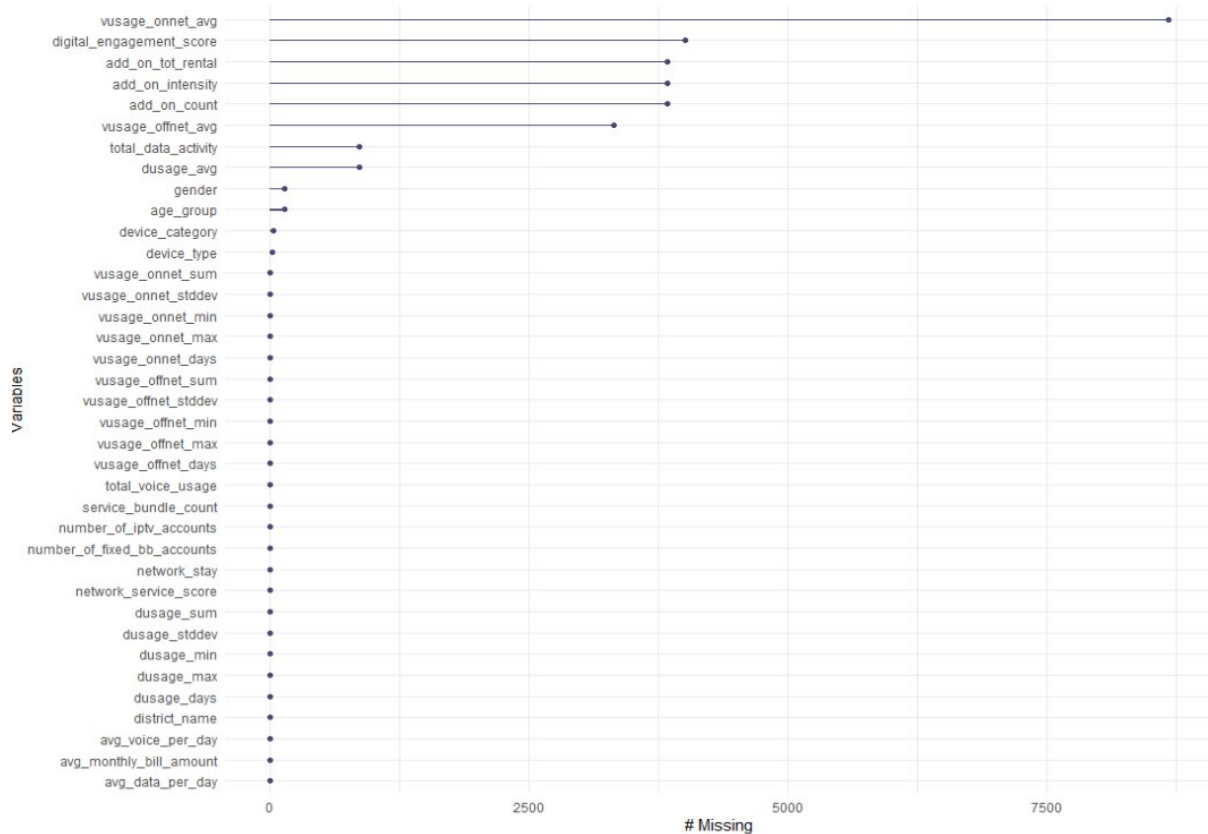


Figure 6: Missing value visualisation.

The missing value visualisation confirmed that most variables have complete data, while a small number of variables contain substantial missing portions. In particular, the variable `vusage_onnet_avg` shows extremely high missingness, suggesting that it may not be reliable for predictive modelling. Understanding the distribution of missing values is important because the treatment method depends on the proportion and importance of missing data (Imtiaz & Shah, 2008).

Q3.2 Methods for handling missing values

Three different missing value handling techniques were applied to evaluate their impact on the dataset:

Method 1: Listwise deletion (removal of missing observations)

The first approach involved removing all observations containing missing values. This method ensures that only complete records are used for modelling (Myers, 2011). The main advantage of this approach is that it produces a clean dataset with no missing data and avoids introducing artificial values. However, the major limitation is that it reduces the dataset size and may remove useful information (Myers, 2011). The results showed that removing missing values changed the distribution of the target variable. Specifically, the original dataset with the mean monthly bill around 2084, after removal is around 2414.

```

> summary(connecttel$avg_monthly_bill_amount)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   -863   1351   2072   2414   2993   15692
>
> summary(connecttel$avg_monthly_bill_amount)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -11446   1036   1784   2084   2701   19008

```

Figure 7: Listwise deletion outcome.

This suggests that removing missing observations may bias the dataset toward higher billing customers. Therefore, this method may distort the underlying data distribution.

Method 2: Median imputation

The second approach involved replacing missing values with the median of each variable. Median imputation is often preferred over mean imputation because it is more robust to extreme values. The results showed that median imputation preserved the distribution of the target variable (Hadeed et al., 2020). Specifically, the original mean is around 2083.9 and after the median imputation, it has become 2083.9, which mean the standard deviation remain unchanged.

```
> summary(data_mean$avg_monthly_bill_amount)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-11446   1036   1784   2084   2701   19008
>
> summary(connecttel$avg_monthly_bill_amount)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-11446   1036   1784   2084   2701   19008
```

Figure 8: Median Imputation outcome.

This suggests that median imputation preserved the statistical properties of the dataset while eliminating missing values. However, this method is particularly suitable when variables exhibit skewed distributions, as seen in the usage variables (Hadeed et al., 2020).

Method 3: K-Nearest Neighbour (KNN) imputation

The third approach involved KNN imputation, which replaces missing values using values from similar observations. This method is considered more advanced because it preserves relationships between variables rather than simply inserting central values (Murti et al., 2019). The results showed that KNN imputation also preserved the distribution. Specifically, the mean remained 2084 which mean the variance remain stable.

```
> summary(data_knn$avg_monthly_bill_amount)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-11446   1036   1784   2084   2701   19008
```

Figure 9: The KNN Imputation outcome.

This indicates that KNN successfully maintained the original structure of the dataset. However, KNN imputation is computationally more expensive and may introduce noise if similarity relationships are weak (Murti et al., 2019).

Q3.3 Comparison of missing value treatments

A comparison of the three techniques revealed important differences. Listwise deletion produced noticeable changes in the target distribution, suggesting that removing observations may introduce bias. Since approximately one-third of some variables contain missing data, removing observations may significantly reduce dataset size and predictive power. In contrast, both median imputation and KNN imputation preserved the original statistical characteristics of the dataset. Density plots showed that the distributions of the imputed datasets closely match the original distribution. Variance comparison also confirmed stability of original variance, median variance and KNN imputation is around 2,162,518. This indicates that both imputation techniques-maintained dataset variability.

```
> variance_compare
      Method Variance
1 Original  2162518
2 Remove   2470462
3 Median   2162518
4 KNN      2162518
```

Figure 10: Variance comparison for 3 methods.

Q3.4 Recommended missing value treatment

Based on the analysis, median imputation is recommended as the most suitable missing value treatment for this dataset. First, median imputation preserved the statistical properties of the dataset while removing missing values. Second, it is robust to outliers, which are common in usage data. Third, it is computationally efficient compared to KNN imputation.

Although KNN imputation also performed well, it introduces additional computational complexity and may not significantly improve model performance compared to median imputation. Additionally, variables with extremely high missing proportions such as **vsusage_onnet_avg** (over 80%) should be considered for removal rather than imputation, as imputing such large portions may introduce artificial patterns.

From a modelling perspective, median imputation is preferred because:

- It maintains distribution stability
- It avoids bias caused by removal
- It improves model usability
- It reduces computational complexity

Therefore, median imputation combined with removal of extremely incomplete variables represents the most practical approach for this dataset. The missing data analysis revealed that while most variables contain complete information, several variables contain substantial missing values. Three treatment approaches were evaluated, including listwise deletion, median imputation, and KNN imputation. The results demonstrated that removing missing observations may distort the dataset, while both median and KNN imputation preserved statistical characteristics. Median imputation was identified as the most suitable approach due to its robustness, simplicity, and ability to maintain dataset integrity. Proper handling of missing data ensures that predictive models developed in later stages will produce reliable and unbiased results.

Q4. Feature Selection and Relationships

Q4.1 Correlation analysis between variables

To identify relationships between variables and reduce redundancy before modelling, a correlation analysis was conducted on all numerical variables using a Pearson correlation matrix (Gravier et al., 2008). The results were visualised using a heatmap, which provided a clear overview of the strength and direction of relationships between variables.

The correlation heatmap revealed several clusters of highly correlated variables, particularly among the data usage metrics and voice usage metrics. For example, variables such as **dsusage_sum**, **total_data_activity**, **avg_data_per_day**, and **dsusage_avg** showed strong positive correlations with each

other. This is expected because these variables measure similar aspects of customer data consumption behaviour.

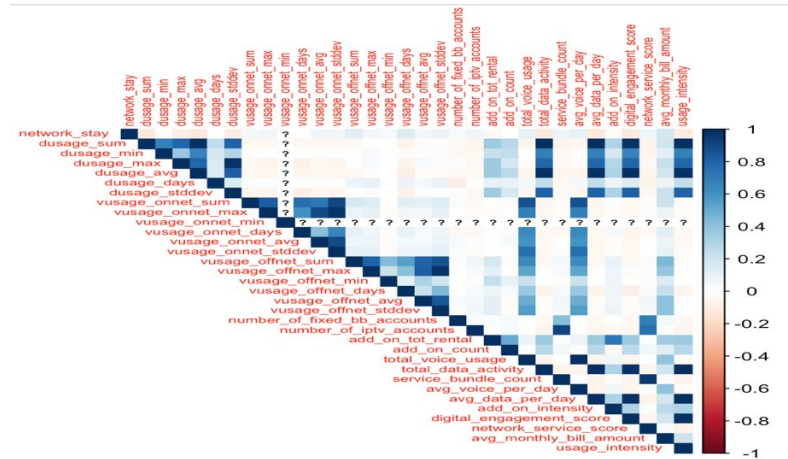


Figure 11: Correlation heatmap.

Similarly, several voice usage variables such as `vusage_offnet_sum`, `total_voice_usage`, and `vusage_offnet_avg` also demonstrated strong correlations, indicating they capture overlapping information about customer calling behaviour. The analysis also showed that the target variable, `avg_monthly_bill_amount`, has moderate positive correlations with several usage-related variables. The strongest correlations were observed with:

- `dusage_sum` ($r \approx 0.389$)
- `avg_data_per_day` ($r \approx 0.389$)
- `usage_intensity` ($r \approx 0.386$)
- `total_data_activity` ($r \approx 0.383$)
- `add_on_tot_rental` ($r \approx 0.381$)

These results indicate that customers with higher service usage and additional service purchases tend to have higher monthly bills.

Q4.2 Dimensionality reduction and removal of redundant variables

To improve model stability and avoid multicollinearity issues, highly correlated variables were identified using a correlation threshold of 0.85 (Kim, 2019). Variables exceeding this threshold were considered redundant and candidates for removal. The variables identified as having potential outliers include `avg_data_per_day`, `dusage_sum`, `digital_engagement_score`, `total_data_activity`, `dusage_avg`, `total_voice_usage`, `dusage_stddev`, `vusage_offnet_sum`, `vusage_offnet_stddev`, `vusage_onnet_stddev`, `vusage_onnet_max`, `network_service_score`, and `number_of_ipv_accounts`.

> `high_cor`

```
[1] "avg_data_per_day"      "dusage_sum"           "digital_engagement_score"
[4] "total_data_activity"   "dusage_avg"           "total_voice_usage"
[7] "dusage_stddev"        "vusage_offnet_sum"    "vusage_offnet_stddev"
[10] "vusage_onnet_stddev"   "vusage_onnet_max"     "network_service_score"
[13] "number_of_ipv_accounts"
```

Figure 12: High Correlation list.

Since these variables capture similar behavioural patterns, retaining all of them would introduce redundancy and potentially distort model estimates. Therefore, dimensionality reduction was performed by removing selected redundant predictors while retaining representative variables. As a result, the number of numeric predictors was reduced from 33 variables to 20 variables, which simplifies the dataset and improves modelling efficiency.

```
> ncol(numeric_data)
[1] 33
>
> ncol(data_reduced)
[1] 20
```

Figure 13: Highly correlated variables removed.

Further confirmation of redundancy was obtained using Variance Inflation Factor (VIF) testing. Extremely high VIF values were observed for:

- total_data_activity (VIF $\approx$ 16143)
- dusage_sum (VIF $\approx$ 16142)

```
> vif(model_check)
total_data_activity    total_voice_usage      dusage_sum    network_stay
           16143.010650             1.011272    16141.913414             1.013508
```

Figure 14: VIF checking.

These values strongly indicate severe multicollinearity. Since both variables measure total data consumption, keeping both would negatively affect regression stability. Therefore, only one representative variable should be retained during modelling. This dimensionality reduction process improves model interpretability, reduces noise, and enhances predictive reliability (Kim, 2019).

Q4.3 Relationships between selected features and the target variable

After removing redundant variables, the relationships between key predictors and the target variable were further explored using scatterplots with fitted regression lines. The visualisations confirmed positive linear relationships between several predictors and the monthly bill amount.

For example:

Total data activity

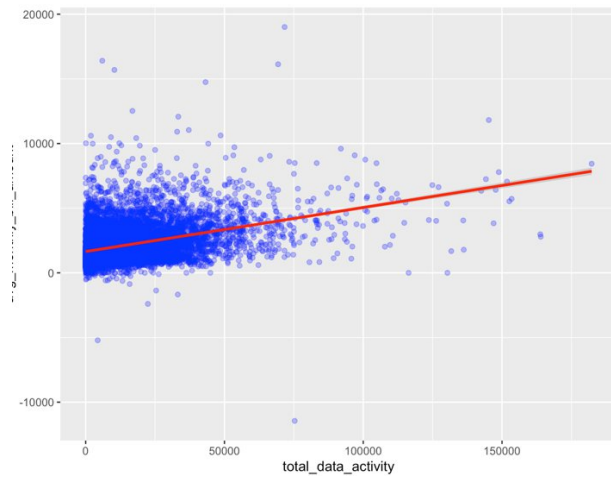


Figure 15: total_data_activity Scatterplot.

The scatterplot shows a clear upward trend, indicating that customers who consume more data tend to have higher monthly bills. Although some variability exists, the positive slope suggests a meaningful relationship between service usage and billing.

Total data usage (dusage_sum)

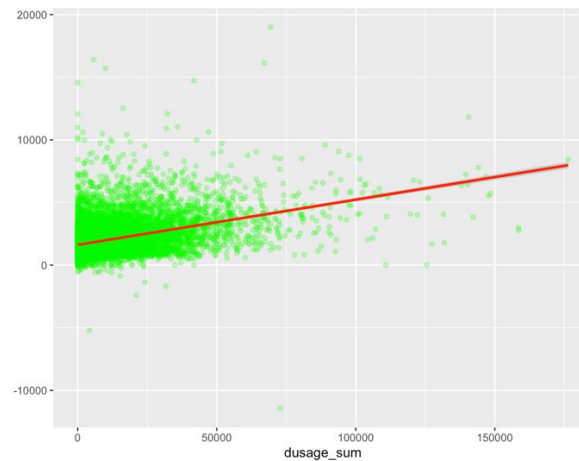


Figure 16: dusage_sum Scatterplot.

A similar pattern is observed, reinforcing the conclusion that higher data consumption leads to higher billing charges. However, since this variable overlaps strongly with total_data_activity, only one should be used in modelling.

Total voice usage

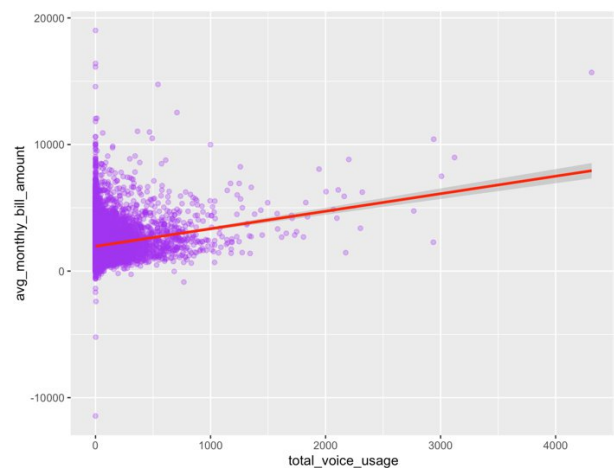


Figure 17: *total_voice_usage* Scatterplot.

Voice usage also shows a positive relationship with billing, although the relationship appears weaker compared to data usage variables. This suggests voice usage contributes to billing but may not be the primary driver. Overall, the visual evidence supports the statistical correlation results and confirms that usage-related variables are important predictors of billing behaviour.

4.4 Key predictors influencing monthly billing

Based on the correlation analysis, multicollinearity testing, and visual inspection, three variables were identified as the most influential predictors of monthly bill amount.

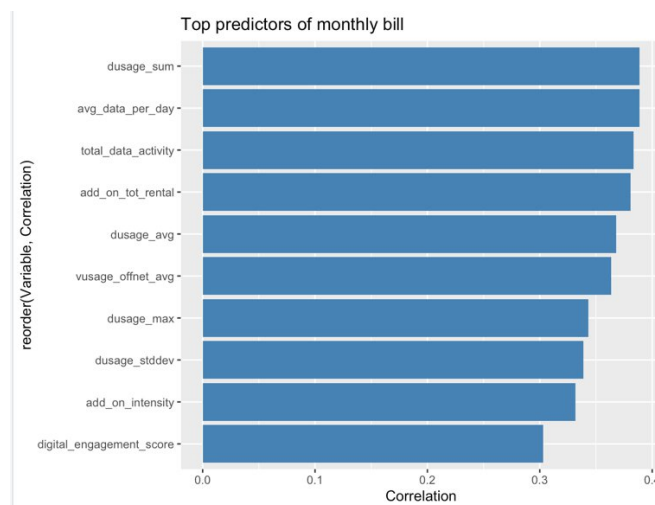


Figure 18: *Top predictors of monthly bill.*

1. Total data activity

This variable shows a strong positive correlation with the monthly bill and represents the overall volume of customer data usage. From a business perspective, data consumption directly contributes to billing charges, making it a logically strong predictor.

2. Add-on total rental

This variable captures the cost of optional services purchased by customers. Its strong positive relationship with billing is expected because additional services directly increase the total monthly charge. This

variable is particularly valuable because it represents customer purchasing behaviour rather than usage alone.

3. Usage intensity

Usage intensity combines multiple usage behaviours into a single metric, reflecting overall customer engagement with telecom services. The strong correlation (approximately 0.386) indicates that customers who use services more intensively tend to incur higher bills.

Overall, the feature selection process demonstrated that customer service usage behaviour is the primary driver of monthly billing. Data consumption variables, add-on purchases, and overall usage intensity were identified as the most important predictors. By removing redundant variables and retaining only representative predictors, the dataset was simplified while preserving the most meaningful information. This process improves model performance by reducing multicollinearity, improving interpretability, and ensuring that the predictive models developed in later stages are both statistically sound and business relevant (Kim, 2019).

Assignment Part 2: Building Predictive Models

Q1. Regression Modelling for Customer Bill Prediction

Q1.1 Development and evaluation of the regression model

To predict customer monthly billing amounts, a multiple linear regression model was developed using the selected predictors identified during the feature selection stage. The selected variables included `total_data_activity`, `add_on_tot_rental`, `usage_intensity`, `network_stay`, and `vusage_offnet_avg`, as these variables showed meaningful relationships with the target variable and represented key aspects of customer service usage behaviour.

The regression model produced the following overall performance results:

- **R-squared = 0.312**
- **Adjusted R-squared = 0.299**
- **RMSE = 1220**
- **MAE = 852**
- **Residual standard error = 1231**

```
Residual standard error: 1231 on 7346 degrees of freedom
Multiple R-squared:  0.2994,    Adjusted R-squared:  0.2989
F-statistic: 627.9 on 5 and 7346 DF,  p-value: < 2.2e-16
```

```
> performance_lm
      Metric      Value
1      RMSE 1220.0294746
2       MAE  852.3009243
3 R-squared   0.3124677
> |
```

Figure 19: The Regression Modelling Performance.

The R-squared value indicates that approximately 31.2% of the variation in monthly billing amounts can be explained by the selected predictors. While this is not extremely high, it is reasonable for customer behaviour data where billing is influenced by many external factors not captured in the dataset. The model F-statistic ($F = 627.9$, $p < 0.001$) indicates that the regression model is statistically significant overall, meaning the predictors collectively contribute to explaining the variation in billing amounts (Schneider et al., 2010). The RMSE value of 1220 indicates the typical prediction error magnitude, while the MAE value of 852 suggests the average absolute deviation between predicted and actual bills. These values indicate moderate predictive accuracy (Schneider et al., 2010).

```
> head(comparison_lm)
```

	Actual	Predicted
1	1977.19	1586.865
2	2925.00	2219.948
3	473.59	1686.132
4	496.34	1843.594
5	2758.85	2477.094
6	2310.88	2662.707

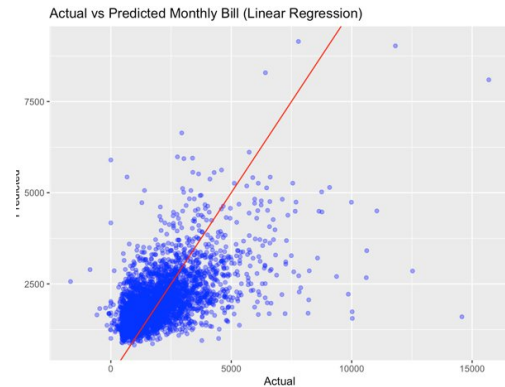


Figure 20: The comparison between the actual vs predicted plot.

The Actual vs Predicted plot further confirms that the model captures the general upward trend between predicted and actual billing values, although some dispersion remains, particularly for customers with very high bills.

Q1.2 Regression model formula and explanation

Based on the regression output, the estimated regression equation is:

$$\text{avg_monthly_bill_amount} = 733.05 - 0.0143(\text{total_data_activity}) + 0.6926(\text{add_on_tot_rental}) + 0.0414(\text{usage_intensity}) + 4.2519(\text{network_stay}) + 48.6149(\text{vusage_offnet_avg})$$

Each component of the model has a specific meaning:

Intercept (733.05)

This represents the baseline predicted monthly bill when all predictors are equal to zero. While this situation may not occur in practice, the intercept provides a starting reference point for the model.

Regression coefficients

Each coefficient represents the expected change in the predicted monthly bill for a one-unit increase in the predictor, assuming other variables remain constant.

For example:

- A one-unit increase in add-on rental increases the predicted bill.
- A one-unit increase in off-network voice usage increases billing.
- Higher usage intensity increases predicted costs.

All predictors were statistically significant ($p < 0.05$), indicating they meaningfully contribute to predicting billing behaviour.

Q1.3 Interpretation of selected predictors

Two important predictors were selected for detailed interpretation: `add_on_tot_rental` and `vusage_offnet_avg`.

Add-on total rental

The coefficient for `add_on_tot_rental` is 0.6926, which means that for every additional unit increase in add-on rental cost, the predicted monthly bill increases by approximately 0.69 units, holding other variables constant. This result is consistent with business expectations because customers who purchase additional services such as premium features or bundled services naturally incur higher monthly charges.

Average off-network voice usage

The coefficient for `vusage_offnet_avg` is 48.61, meaning that a one-unit increase in average off-network voice usage is associated with an increase of approximately 48.6 units in the predicted monthly bill, assuming other variables remain constant. This suggests that off-network calls may be priced higher or incur additional charges compared to on-network calls, making this a strong billing driver. These interpretations demonstrate that both service usage behaviour and optional service purchases play important roles in determining billing outcomes.

Q1.4 Reliability and limitations of the regression model

The regression model shows reasonable reliability but also presents several limitations.

Strengths of the model

The model demonstrates several positive characteristics. First, all predictors are statistically significant, indicating they meaningfully contribute to explaining billing variation. (Heyer & Stamm, 2013) Second, the overall F-test confirms that the model provides a better fit than a model without predictors. Third, the diagnostic plots suggest the model captures the general trend between predictors and billing values (Heyer & Stamm, 2013).

Limitations of the model

Despite these strengths, several limitations were identified.

- **Moderate explanatory power**
The R-squared value of 0.312 indicates that a large proportion of variation in billing remains unexplained. This suggests other factors such as pricing plans, promotions, or customer demographics may also influence billing but were not included.
- **Presence of heteroscedasticity**
The Scale-Location plot suggests increasing variance in residuals as fitted values increase, which indicates potential heteroscedasticity. This may reduce prediction reliability for high-value customers (Heyer & Stamm, 2013).
- **Multicollinearity concerns**
Variance Inflation Factor results show high VIF values for `total_data_activity` and `usage_intensity` (around 40), indicating some remaining multicollinearity. This occurs because both variables measure related usage behaviour. While still usable, this may affect coefficient stability.
- **Outliers and extreme values**
The residual plots show some extreme observations, which may influence model accuracy. Customers with unusually high or low bills may not be well represented by a linear model (Heyer & Stamm, 2013).

Potential improvements

The model could be improved by:

- Applying log transformation to skewed variables
- Removing or treating extreme outliers
- Testing interaction effects between usage variables
- Comparing with non-linear models such as decision trees

Overall, the regression model provides a statistically significant and reasonably interpretable representation of the relationship between customer usage behaviour and monthly billing. The results confirm that service usage, add-on purchases, and voice consumption are key drivers of customer billing outcomes. However, the moderate explanatory power and presence of some modelling limitations suggest that additional modelling techniques, such as decision trees, may further improve prediction accuracy.

Q2. Decision Tree Modelling for Customer Bill Prediction

Q2.1 Construction of the decision tree model

A decision tree regression model was developed to predict customer monthly bill amounts using the selected predictors identified in the feature selection stage (Myles et al., 2004). The model was constructed using the rpart algorithm with the ANOVA method, which is appropriate for continuous target variables (Wang et al., 2020). The predictors used in the model will be total_data_activity, add_on_tot_rental, usage_intensity, network_stay and vusage_offnet_avg. These variables were selected because they represent key customer behaviour factors such as service consumption, add-on purchases, customer engagement, and service tenure.

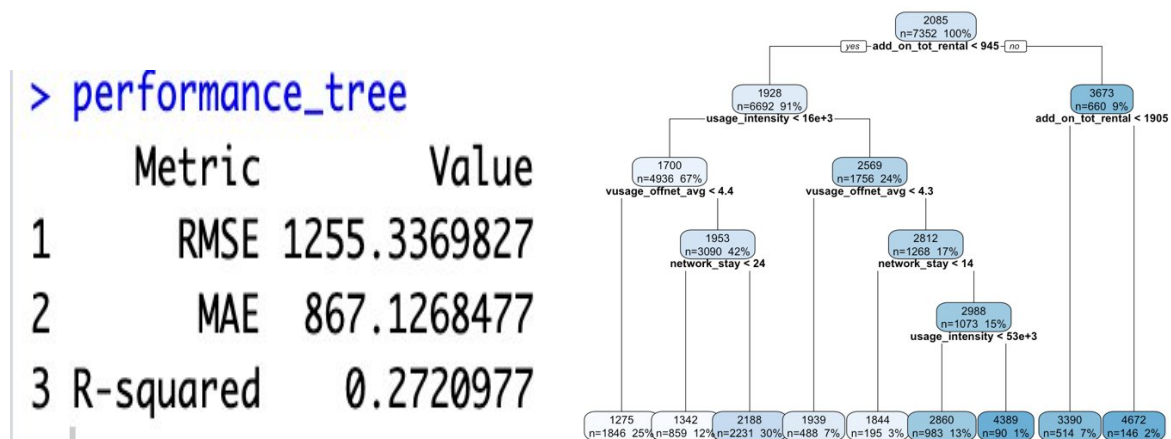


Figure 21: The Decision Tree Model and The Performance.

The decision tree achieved the following performance results:

- **RMSE = 1255**
- **MAE = 867**
- **$R^2 = 0.272$**

These results indicate that the decision tree explains approximately 27.2% of the variation in monthly bills. While slightly lower than the regression model, the decision tree provides useful interpretability advantages because it clearly shows how customer characteristics influence billing decisions.

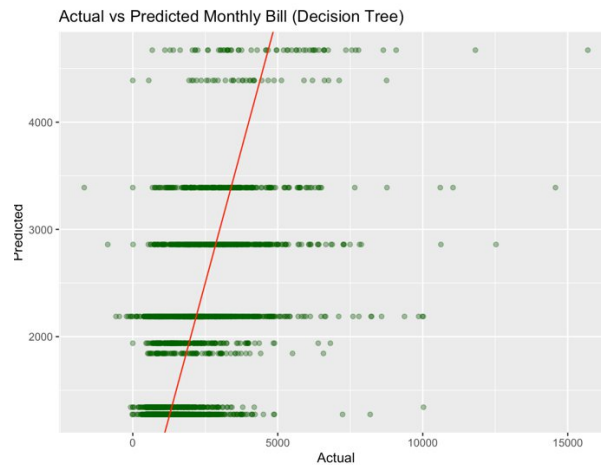


Figure 22: The Actual vs Predicted Plot.

The Actual vs Predicted plot shows that the tree captures general billing patterns but produces grouped predictions, which is typical for decision trees because predictions are based on terminal node averages rather than continuous equations.

Q2.2 Examination of tree structure and decision paths

The decision tree structure shows that the first and most important split occurs on `add_on_tot_rental`, indicating that add-on service costs are the primary driver of billing differences. The first major split divides customers into those with `add_on_tot_rental < 945`, who generally have lower predicted bills and are further segmented based on `usage_intensity` and `voice_usage` behaviour, and those with `add_on_tot_rental ≥ 945`, who tend to have higher predicted bills, reflecting the strong impact of optional services on monthly costs. Further important decision nodes include `usage_intensity`, where customers with lower usage intensity tend to have lower bills while those with higher usage intensity are associated with higher predicted bills, and `vusage_offnet_avg`, where customers with higher off-network voice usage tend to have higher billing levels, suggesting additional charges from off-network calls. Another contributing factor is `network_stay`, where customers with longer tenure sometimes fall into higher billing groups, possibly reflecting service upgrades or bundled services over time. Overall, the tree structure suggests that billing is primarily driven by service purchases first, followed by usage behaviour and customer tenure.

Q2.3 Feature importance analysis

Feature importance analysis shows the relative contribution of each variable in predicting billing outcomes.

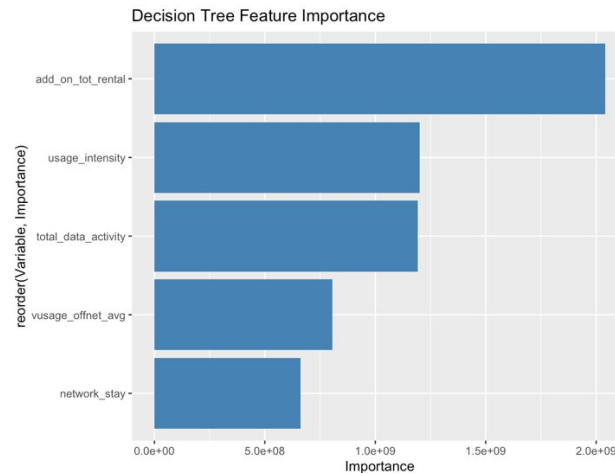


Figure 23: Decision Tree Feature Importance.

The importance ranking was:

1. add_on_tot_rental
2. usage_intensity
3. total_data_activity
4. vusage_offnet_avg
5. network_stay

The results indicate that add_on_tot_rental is the most influential variable, confirming that optional service purchases are the strongest determinant of billing levels. Usage intensity and total data activity were also highly influential, confirming that customer consumption behaviour is a key billing driver. Network stay had the lowest importance, suggesting that customer tenure plays a smaller role compared to usage and purchasing behaviour. These results are consistent with earlier correlation findings and strengthen the reliability of the feature selection process.

Q2.4 Hyperparameter experimentation

To improve model performance, several decision tree configurations were tested using different hyperparameter settings, including tree depth, minimum split size, and complexity parameters (Mantovani et al., 2024).

Three tree configurations were compared:

```
> tree_comparison
      Model      RMSE      MAE R_squared
1 Shallow Tree 1291.546 903.3158 0.2295013
2 Medium Tree 1242.890 855.6302 0.2864610
3 Deep Tree 1283.291 841.1936 0.2393191
```

Figure 24: Compare tuned trees.

The medium tree configuration performed best, achieving the lowest RMSE and highest R^2 value. This indicates that moderate tree complexity provides the best balance between underfitting and overfitting. The shallow tree underperformed because it was too simple to capture important customer behaviour patterns.

The deep tree showed slightly worse generalisation, suggesting potential overfitting. Pruning was also applied using the optimal complexity parameter to further improve model generalisation and reduce unnecessary splits.

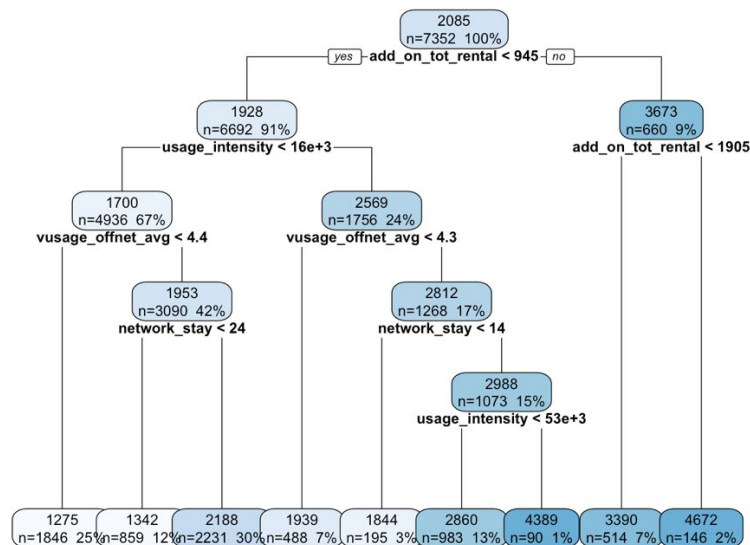


Figure 25: The Pruned Tree.

This tuning process demonstrates good modelling practice by ensuring the tree is neither too simple nor overly complex.

Q2.5 Human-readable decision rules and example prediction

One advantage of decision trees is their interpretability, which allows the model to be translated into human-readable rules (Berge et al., 2019).

Example rules extracted from the tree include:

Rule example 1:

If:

- $\text{add_on_tot_rental} < 945$
- $\text{usage_intensity} < 16117$
- $\text{vusage_offnet_avg} < 4.4$

Then predicted monthly bill $\approx$ **1275**

Rule example 2:

If:

- $\text{add_on_tot_rental} < 945$
- $\text{usage_intensity} \geq 16117$
- $\text{vusage_offnet_avg} \geq 4.3$
- $\text{network_stay} \geq 14$

Then predicted monthly bill $\approx$ **2860**

Rule example 3:

If:

- $\text{add_on_tot_rental} \geq 1905$

Then predicted monthly bill $\approx$ **4672**

These rules clearly show how the model segments customers into billing categories based on service purchases and usage behaviour.

Example observation path

For example, one selected observation had the following characteristics:

- $\text{total_data_activity} = 15274$
- $\text{add_on_tot_rental} = 110$
- $\text{usage_intensity} = 15274$
- $\text{network_stay} = 35$
- $\text{vusage_offnet_avg} = 4.42$

The decision tree processes this observation as follows:

Step 1:

$\text{add_on_tot_rental} < 945 \rightarrow$ move left branch

Step 2:

$\text{usage_intensity} < 16117 \rightarrow$ move left branch

Step 3:

$\text{vusage_offnet_avg} \geq 4.4 \rightarrow$ move right branch

Step 4:

$\text{network_stay} \geq 24 \rightarrow$ move right branch

Final prediction:

Predicted monthly bill $\approx$ **2188**

The actual bill for this customer was approximately 1977, showing that the model provides a reasonable estimate. Overall, the decision tree model provides a clear and interpretable representation of how customer behaviour influences billing. The results confirm that add-on services, usage behaviour, and customer engagement are the primary drivers of billing outcomes. Although the decision tree showed slightly lower predictive accuracy compared to the regression model, it provides valuable business insights due to its transparency and ability to generate understandable decision rules. Together with the regression model, the decision tree provides complementary perspectives on customer billing behaviour and supports evidence-based business decision making (Berge et al., 2019).

Q3. Model Comparison and Business Recommendation

Q3.1 Importance of experimenting with different model configurations and hyperparameters

Experimenting with different model configurations and hyperparameters is a critical step in predictive analytics because model performance can vary significantly depending on how the model is structured. A

single model configuration rarely provides the best performance, and tuning parameters allows analysts to find the best balance between accuracy, generalisation ability, and interpretability. For example, in regression models, feature selection affects how well the model explains the target variable. Including too many variables may introduce noise and multicollinearity, while excluding important predictors may reduce predictive accuracy. Similarly, in decision tree models, hyperparameters such as tree depth, minimum samples per split, and complexity parameters influence whether the model underfits or overfits the data. Testing different configurations also helps ensure that the model generalises well to unseen data rather than simply memorising the training data. This improves model robustness and reliability, which is particularly important in real-world business environments where models must perform consistently across different customer segments.

Therefore, experimenting with model configurations is necessary to:

- Improve predictive accuracy
- Reduce overfitting
- Improve model stability
- Ensure practical usability of the model

This process ultimately helps identify the most appropriate model for both technical performance and business application.

Q3.2 Comparison of regression and decision tree models

The linear regression and decision tree models were compared using predictive accuracy, interpretability, robustness, and suitability for business decision-making.

Predictive accuracy

The regression model demonstrated slightly better predictive performance compared to the decision tree model.

```
> model_comparison
      Model    RMSE    MAE R_squared
1 Linear Regression 1220.03 852.30    0.312
2   Decision Tree 1255.34 867.12    0.272
```

Figure 26: Model Comparison.

The regression model achieved a lower RMSE and MAE, indicating smaller prediction errors on average. Additionally, the regression model achieved a higher R-squared value (0.312 vs 0.272), suggesting it explains more variation in customer monthly bills.

Although the performance difference is not extremely large, the regression model shows consistently better predictive accuracy.

Interpretability

Both models are interpretable, but in different ways. The regression model provides clear mathematical relationships between predictors and the target variable through coefficients. This allows analysts to understand how much each variable contributes to changes in the monthly bill.

For example:

- Add-on rental increases the bill directly

- Network stay duration increases cost gradually
- Usage intensity increases expected monthly spending

However, decision trees provide more intuitive interpretation because they translate predictions into simple rules. For example:

If add-on rental < 945

AND usage intensity < threshold

THEN predicted bill $\approx$ lower range

These rule-based structures are easier for non-technical stakeholders to understand.

Therefore:

- Regression $\rightarrow$ better statistical interpretability
- Decision Tree $\rightarrow$ better business interpretability

Robustness

Regression models tend to be more stable when relationships between variables are approximately linear. However, they may be sensitive to multicollinearity and outliers. Decision trees are more flexible because they can capture non-linear relationships automatically. However, they are more sensitive to data variation and may change structure significantly with small data changes unless properly pruned (Apté & Weiss, 1997). In this analysis, the regression model showed slightly better stability based on performance consistency and lower error variation. The decision tree showed slightly lower R-squared, suggesting weaker generalisation performance.

Therefore:

- Regression $\rightarrow$ more stable performance
- Decision Tree $\rightarrow$ more flexible but slightly less robust

Suitability for practical decision-making

From a business perspective, both models offer different advantages.

Regression models are useful for:

- Budget forecasting
- Revenue prediction
- Cost optimisation analysis
- Understanding financial drivers

Decision trees are useful for:

- Customer segmentation
- Identifying high-value customers
- Explaining pricing differences
- Supporting operational decisions

In the ConnectTel context, where predicting monthly bills is important for revenue planning, the regression model provides stronger financial prediction capability. However, the decision tree can support marketing and customer management decisions by identifying behavioural segments.

Q3.3 Trade-off between model simplicity and predictive performance

When selecting predictive models, there is often a trade-off between simplicity and predictive performance. Simpler models such as linear regression are easier to interpret, faster to implement, and easier to maintain. They also reduce the risk of overfitting when the relationships between variables are relatively straightforward. More complex models, such as deeper decision trees or ensemble models, may capture more complex patterns but may sacrifice interpretability and increase maintenance complexity. In this study, the regression model is simpler while also providing slightly better predictive accuracy. This is an ideal situation because normally improving performance requires increasing complexity. Here, regression achieves both simplicity and better performance compared to the decision tree. This suggests that the underlying relationships between customer usage behaviour and monthly bills are largely linear, making regression an appropriate modelling choice.

From a business implementation perspective, simpler models are often preferred because they:

- Require less computational resources
- Are easier to explain to stakeholders
- Are easier to monitor
- Reduce operational risk

Therefore, unless a complex model significantly improves prediction accuracy, simpler models are generally preferred.

Q3.4 Recommended modelling approach for ConnectTel

Based on the analysis, the linear regression model is recommended as the primary predictive model for ConnectTel.

From a technical perspective, the regression model demonstrates:

- Lower RMSE (better prediction accuracy)
- Lower MAE (smaller average errors)
- Higher R-squared (better explanatory power)
- Greater model stability

These characteristics indicate that regression provides more reliable predictions for customer billing. From a business perspective, regression also provides clear financial insights. The model coefficients allow ConnectTel to understand how factors such as add-on services, network usage, and data activity influence revenue. This supports pricing strategy, customer targeting, and revenue forecasting. However, the decision tree model should not be discarded. Instead, it can be used as a complementary analytical tool to:

- Identify customer segments
- Support marketing strategies
- Detect high-value usage patterns
- Improve customer retention strategies

Therefore, the recommended approach is:

Primary model:

Linear regression for bill prediction and financial planning.

Supporting model:

Decision tree for customer behaviour insights and segmentation.

This combined approach allows ConnectTel to benefit from both predictive accuracy and business interpretability, supporting both operational and strategic decision-making.

References

- Alexandropoulos, S. N., Kotsiantis, S. B., & Vrahatis, M. N. (2019). Data preprocessing in predictive data mining. *The Knowledge Engineering Review*, 34. <https://doi.org/10.1017/s026988891800036x>
- Anderson, J. A. (1984). Regression and ordered categorical variables. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 46(1), 1–22. <https://doi.org/10.1111/j.2517-6161.1984.tb01270.x>
- Apté, C., & Weiss, S. (1997). Data mining with decision trees and decision rules. *Future Generation Computer Systems*, 13(2–3), 197–210. [https://doi.org/10.1016/s0167-739x\(97\)00021-6](https://doi.org/10.1016/s0167-739x(97)00021-6)
- Baladram, S. (2025, January 13). *Encoding Categorical Data, Explained: A Visual Guide with Code Example for Beginners*. Towards Data Science. <https://towardsdatascience.com/encoding-categorical-data-explained-a-visual-guide-with-code-example-for-beginners-b169ac4193ae/>
- Berge, G. T., Granmo, O., Tveit, T. O., Goodwin, M., Jiao, L., & Matheussen, B. V. (2019). Using the TseTlin machine to learn Human-Interpretable Rules for High-Accuracy Text Categorization with medical applications. *IEEE Access*, 7, 115134–115146. <https://doi.org/10.1109/access.2019.2935416>
- Biecek, P., & Burzykowski, T. (2021). *Explanatory model analysis*. <https://doi.org/10.1201/9780429027192>
- Czajkowski, M., & Kretowski, M. (2016). The role of decision tree representation in regression problems – An evolutionary perspective. *Applied Soft Computing*, 48, 458–475. <https://doi.org/10.1016/j.asoc.2016.07.007>
- Emmanuel, T., Maupong, T., Mpoeleng, D., Semong, T., Mphago, B., & Tabona, O. (2021a). A survey on missing data in machine learning. *Journal of Big Data*, 8(1), 140. <https://doi.org/10.1186/s40537-021-00516-9>
- Emmanuel, T., Maupong, T., Mpoeleng, D., Semong, T., Mphago, B., & Tabona, O. (2021b). A survey on missing data in machine learning. *Journal of Big Data*, 8(1), 140. <https://doi.org/10.1186/s40537-021-00516-9>

Fenlon, C., O'Grady, L., Doherty, M. L., & Dunnion, J. (2017). A discussion of calibration techniques for evaluating binary and categorical predictive models. *Preventive Veterinary Medicine*, 149, 107–114. <https://doi.org/10.1016/j.prevetmed.2017.11.018>

Frank, E., Wang, Y., Inglis, S., Holmes, G., & Witten, I. H. (1998). Using model trees for classification. *Machine Learning*, 32(1), 63–76. <https://doi.org/10.1023/a:1007421302149>

Gravier, J., Vignal, V., Bissey-Breton, S., & Farre, J. (2008). The use of linear regression methods and Pearson's correlation matrix to identify mechanical–physical–chemical parameters controlling the micro-electrochemical behaviour of machined copper. *Corrosion Science*, 50(10), 2885–2894. <https://doi.org/10.1016/j.corsci.2008.07.022>

Hadeed, S. J., O'Rourke, M. K., Burgess, J. L., Harris, R. B., & Canales, R. A. (2020). Imputation methods for addressing missing data in short-term monitoring of air pollutants. *The Science of the Total Environment*, 730, 139140. <https://doi.org/10.1016/j.scitotenv.2020.139140>

Heyer, T., & Stamm, J. (2013). Levee reliability analysis using logistic regression models – abilities, limitations and practical considerations. *Georisk Assessment and Management of Risk for Engineered Systems and Geohazards*, 7(2), 77–87. <https://doi.org/10.1080/17499518.2013.790734>

Imtiaz, S. A., & Shah, S. L. (2008). Treatment of missing values in process data analysis. *The Canadian Journal of Chemical Engineering*, 86(5), 838–858. <https://doi.org/10.1002/cjce.20099>

Kalisch, M., Michalak, M., Sikora, M., Wróbel, L., & Przyszałka, P. (2016). Influence of outliers introduction on predictive models quality. In *Communications in computer and information science* (pp. 79–93). https://doi.org/10.1007/978-3-319-34099-9_5

Kim, J. H. (2019). Multicollinearity and misleading statistical results. *Korean Journal of Anesthesiology*, 72(6), 558–569. <https://doi.org/10.4097/kja.19087>

Kumar, S., & Chong, I. (2018). Correlation analysis to identify the effective data in machine learning: prediction of depressive disorder and emotion states. *International Journal of Environmental Research and Public Health*, 15(12), 2907. <https://doi.org/10.3390/ijerph15122907>

Lalla, M. (2016). Fundamental characteristics and statistical analysis of ordinal variables: a review. *Quality & Quantity*, 51(1), 435–458. <https://doi.org/10.1007/s11135-016-0314-5>

Loh, W. (2011). Classification and regression trees. *Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery*, 1(1), 14–23. <https://doi.org/10.1002/widm.8>

Mantovani, R. G., Horváth, T., Rossi, A. L. D., Cerri, R., Barbon, S., Junior, Vanschoren, J., & De Carvalho, A. C. P. L. F. (2024). Better trees: an empirical study on hyperparameter tuning of classification decision tree induction algorithms. *Data Mining and Knowledge Discovery*, 38(3), 1364–1416.
<https://doi.org/10.1007/s10618-024-01002-5>

McCarthy, R. V., McCarthy, M. M., & Ceccucci, W. (2022). Introduction to predictive Analytics. In *Introduction to Predictive Analysis* (pp. 1–26). https://doi.org/10.1007/978-3-030-83070-0_1

Modelling Binary Data, Second Edition. (n.d.). Google Books.
https://books.google.com/books?hl=vi&lr=&id=3Zx_AwAAQBAJ&oi=fnd&pg=PP1&dq=Binary+variables+contain+only+two+possible+outcomes+and+are+often+used+to+represent+presence+or+absence+of+a+characteristic.+&ots=w5qnM1qljo&sig=zSuDr3i3xI_vcNDWrgndk4mOc2M

Murti, D. M. P., Pujianto, U., Wibawa, A. P., & Akbar, M. I. (2019). K-Nearest Neighbor (K-NN) based Missing Data Imputation. *IEEE Explore*, 83–88. <https://doi.org/10.1109/icsitech46713.2019.8987530>

Myers, T. A. (2011). Goodbye, Listwise deletion: presenting hot deck imputation as an easy and effective tool for handling missing data. *Communication Methods and Measures*, 5(4), 297–310.
<https://doi.org/10.1080/19312458.2011.624490>

Myles, A. J., Feudale, R. N., Liu, Y., Woody, N. A., & Brown, S. D. (2004). An introduction to decision tree modeling. *Journal of Chemometrics*, 18(6), 275–285. <https://doi.org/10.1002/cem.873>

Rakrak, M. (2025). Exploring variability in data: the role of range, variance, and standard deviation. *INTERNATIONAL JOURNAL OF MULTIDISCIPLINARY RESEARCH AND ANALYSIS*, 08(03).
<https://doi.org/10.47191/ijmra/v8-i03-47>

Robitzsch, A. (2020). Why ordinal variables can (Almost) always be treated as continuous variables: Clarifying assumptions of robust continuous and ordinal factor analysis estimation methods. *Frontiers in Education*, 5. <https://doi.org/10.3389/feduc.2020.589965>

Sand Technologies. (2025, December 15). *5 steps to prepare your data for AI*.
<https://www.sandtech.com/insight/5-steps-to-prepare-your-data-for-ai/#:~:text=Collect%20data%20representative%20of%20the,Note%20all%20data%20formats>

Schneider, A., Hommel, G., & Blettner, M. (2010). Linear Regression analysis. *Deutsches Ärzteblatt International*, 107(44), 776–782. <https://doi.org/10.3238/arztebl.2010.0776>

Subex Limited. (n.d.). *SubEX - pioneers in digital trust and business optimisation* [Video]. Subex.
<https://www.subex.com/glossary/customer-segmentation-in-the-telecom->

industry/#:~:text=This%20technique%20divides%20customers%20based,reduce%20churn%20and%20enhance%20profitability.

Uher, J. (2021). Functions of units, scales and quantitative data: Fundamental differences in numerical traceability between sciences. *Quality & Quantity*, 56(4), 2519–2548. <https://doi.org/10.1007/s11135-021-01215-6>

Walelgne, E. A., Asrese, A. S., Manner, J., Bajpai, V., & Ott, J. (2020). Clustering and predicting the data usage patterns of geographically diverse mobile users. *Computer Networks*, 187, 107737. <https://doi.org/10.1016/j.comnet.2020.107737>

Wang, F., Huang, G. H., Fan, Y., & Li, Y. P. (2020). Robust subsampling ANOVA methods for sensitivity analysis of water resource and environmental models. *Water Resources Management*, 34(10), 3199–3217. <https://doi.org/10.1007/s11269-020-02608-2>